

LECTURE 8

GROUP AND RING THEORY

Next, we prove two theorems providing us with criteria for a module to be Noetherian or Artinian.

Theorem 6.5. *Let M be an R -module. The following are equivalent:*

- (i) M is Noetherian;
- (ii) every submodule of M is finitely generated;
- (iii) every non-empty set S of submodules of M has a maximal element (that is, a submodule M_0 in S such that for any submodule N_0 in S with $N_0 \supseteq M_0$ we have $N_0 = M_0$).

Proof. (i) $\implies$ (ii): Let N be a submodule of M . Assume that N is not finitely generated. For any $k \in \mathbb{N}$ let $a_1, \dots, a_k \in N$. Then $N \neq (a_1, \dots, a_k)$. Choose $a_{k+1} \in N$ such that $a_{k+1} \notin (a_1, \dots, a_k)$. We then obtain an infinite properly ascending chain

$$(a_1) \subsetneq (a_1, a_2) \subsetneq \cdots \subsetneq (a_1, \dots, a_k) \subsetneq (a_1, \dots, a_{k+1}) \subsetneq \cdots$$

of submodules of M , which is a contradiction to the hypothesis. Hence N is finitely generated.

(ii) $\implies$ (iii): Let N_0 be an element of S . If N_0 is not maximal, it is properly contained in a submodule $N_1 \in S$. If N_1 is not maximal, then N_1 is properly contained in a submodule $N_2 \in S$. If S has no maximal elements, we obtain an infinite properly ascending chain of submodules

$$N_0 \subset N_1 \subset N_2 \subset \cdots$$

of M . Let $N = \bigcup_i N_i$. Then N is a submodule of M . Indeed, let $x, y \in \bigcup_i N_i$ and $r \in R$. Then $x \in N_j$ and $y \in N_k$ for some $j, k \in \mathbb{N}$. Let $\ell = \max\{j, k\}$. Then

$$x, y \in N_\ell,$$

hence

$$x - y \in N_\ell \quad \text{and} \quad rx \in N_\ell.$$

Thus $x - y \in N$ and $rx \in N$ and hence N is a submodule of M , as claimed. By (ii) we have that N is finitely generated, so there exist elements $a_1, \dots, a_n \in N$, for some $n \in \mathbb{N}$, such that $N = (a_1, \dots, a_n)$. Let u_i be such that $a_i \in N_{u_i}$, and let $\mu := \max\{u_1, \dots, u_n\}$. Then $a_1, \dots, a_n \in N_\mu$. Therefore

$$N = (a_1, \dots, a_n) \subseteq N_\mu \subseteq N$$

which implies that $N_\mu = N$. Then $N_\mu = N_{\mu+1} = \cdots$, a contradiction. Thus S must have a maximal element.

(iii) $\implies$ (i): Let $M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$ be an ascending chain of submodules of M . By (iii), the sequence $M_1, M_2, M_3, \dots$ has a maximal element, say M_k . Then $M_k = M_{k+1} = \cdots$, which gives that M is Noetherian. $\square$

Note that a maximal element does not need to be unique. For instance, if $M = \mathbb{Z}$ as a $\mathbb{Z}$ -module, and if $S = \{4\mathbb{Z}, 2\mathbb{Z}, 9\mathbb{Z}, 3\mathbb{Z}\}$, then we have two maximal elements in S , namely $2\mathbb{Z}$ and $3\mathbb{Z}$.

We now define a concept that is dual to that of a finitely generated module.

Definition 6.6. An R -module M is said to be *finitely cogenerated* if, for each family $(M_\alpha)_{\alpha \in \Lambda}$ of submodules of M ,

$$\bigcap_{\alpha \in \Lambda} M_\alpha = 0 \implies \bigcap_{\alpha \in \Lambda'} M_\alpha = 0$$

for some finite subset Λ' of Λ .

Example 6.7. The cyclic group $\mathbb{Z}/p^n\mathbb{Z}$, for p a prime, considered as a $\mathbb{Z}$ -module is finitely cogenerated. Indeed, the subgroups are

$$(0), \quad \text{and} \quad (p^k) \text{ for } 1 \leq k \leq n-1,$$

and we have

$$(p) \supset (p^2) \supset (p^3) \supset \cdots \supset (p^{n-1}) \supset (0).$$

Let $(M_\alpha)_{\alpha \in \Lambda}$ be a family of submodules of $\mathbb{Z}/p^n\mathbb{Z}$, and suppose $\bigcap_{\alpha \in \Lambda} M_\alpha = 0$. If for all $\alpha \in \Lambda$, we have $M_\alpha \supseteq (p^{n-1})$ then

$$\bigcap_{\alpha \in \Lambda} M_\alpha \supseteq (p^{n-1}) \supsetneq (0).$$

So $\bigcap M_\alpha = (0)$ only if there exists $\beta \in \Lambda$ such that $M_\beta = 0$. Then $\Lambda' = \{\beta\}$.

Note that the infinite-dimensional vector space $R = \mathbb{R}^\infty$, as an R -module, is not finitely cogenerated. Indeed, for each $i \in \mathbb{N}$, writing $M_i = \mathbb{R} \times \cdots \times \mathbb{R} \times \{0\} \times \mathbb{R} \times \cdots$ for the subspace of R consisting of all sequences with zero i th component, we see that $\bigcap_{i \in \mathbb{N}} M_i = 0$ but no intersection of finitely many of the M_i 's is trivial.

The next theorem is dual to the previous.

Theorem 6.8. *Let M be an R -module. The following are equivalent:*

- (i) M is Artinian;
- (ii) every quotient module of M is finitely cogenerated;
- (iii) every non-empty set S of submodules of M has a minimal element (that is, a submodule M_0 in S such that for any submodule N_0 in S with $N_0 \subseteq M_0$ we have $N_0 = M_0$).

Proof. The proof is dual to the proof of Theorem 6.5 and left as an exercise (see Exercise Sheet 9). $\square$

Theorem 6.9. *Let M be a Noetherian (respectively an Artinian) R -module and let $K \subseteq M$ be a submodule. Then K and M/K are Noetherian (respectively Artinian).*

Proof. This follows from Theorem 6.5 and Theorem 4.31, as seen below. From Theorem 6.5, we have that an R -module N is Noetherian if and only if every submodule of N is finitely generated.

As every submodule of K is a submodule of M , it is finitely generated. Hence K is Noetherian.

From Theorem 4.31, every submodule of M/K is of the form L/K where $K \subseteq L \subseteq M$ is a submodule. Thus L is finitely generated and hence L/K is finitely generated. Therefore M/K is Noetherian.

Artinian case: If $K \supseteq L_1 \supseteq L_2 \supseteq \cdots$ is a descending sequence of submodules of K , then this is also a sequence in M , so it must stabilise.

If $M/K \supseteq L_1/K \supseteq L_2/K \supseteq \cdots$ is a descending sequence of submodules of M/K , then we have a descending sequence $M \supseteq L_1 \supseteq L_2 \supseteq \cdots$ of submodules in M , so it stabilises. Hence M/K is Artinian. $\square$

Theorem 6.10. *Let M be an R -module, and let N be an R -submodule of M . Then M is Noetherian (respectively Artinian) if and only if both N and M/N are Noetherian (respectively Artinian).*

Proof. ($\Rightarrow$) This is Theorem 6.9.

($\Leftarrow$) Let N and M/N be Noetherian, and let K be any submodule of M . Then $(K+N)/N$ is a submodule of M/N , and hence it is finitely generated (Theorem 6.5). However then $(K+N)/N \cong K/(N \cap K)$ implies $K/(N \cap K)$ is finitely generated, say

$$\frac{K}{N \cap K} = (\bar{x}_1, \dots, \bar{x}_m), \quad \text{for } \bar{x}_i := x_i + (N \cap K) \in \frac{K}{N \cap K}.$$

Then

$$K = (x_1, \dots, x_m) + N \cap K, \quad \text{for } x_i \in K.$$

Further, as N is Noetherian, its submodule $N \cap K$ is finitely generated, say by $y_1, \dots, y_n \in N \cap K$. This implies that

$$K = (x_1, \dots, x_m) + (y_1, \dots, y_n).$$

Hence K is finitely generated, so M is Noetherian.

The proof for the Artinian case is similar. Let $M \supseteq K_1 \supseteq K_2 \supseteq \dots$ be a descending sequence. As M/N is Artinian, there is an $m \in \mathbb{N}$ such that

$$\frac{K_{m+i} + N}{N} = \frac{K_m + N}{N} \quad \text{for all } i \geq 0,$$

that is

$$K_{m+i} + N = K_m + N \quad \text{for all } i \geq 0.$$

As $(K_1 \cap N) \supseteq (K_2 \cap N) \supseteq \dots$ is a descending sequence in N , and N is Artinian there is an $n \in \mathbb{N}$ such that

$$K_{n+i} \cap N = K_n \cap N \quad \text{for all } i \geq 0.$$

Set $q = \max\{m, n\}$. We claim that

$$K_{q+i} = K_q \quad \text{for all } i \geq 0.$$

Indeed, for $i \geq 0$, it is clear that $K_{q+i} \subseteq K_q$. Let then $x \in K_q$. Since $K_{q+i} + N = K_q + N$, we have $y \in K_{q+i}$ and $u \in N$ such that $y + u = x$. So $u = x - y \in K_q$, since $y \in K_{q+i} \subseteq K_q$. Therefore

$$u \in K_q \cap N = K_{q+i} \cap N,$$

and it follows that

$$x = y + u \in K_{q+i}.$$

We therefore have

$$K_q \subseteq K_{q+i} \subseteq K_q,$$

which yields $K_q = K_{q+i}$, as claimed. Thus M is Artinian.

[Here is an alternative proof for the converse direction. Consider an ascending chain of submodules

$$0 \subseteq M_1 \subseteq M_2 \subseteq \dots$$

in M . Consider the intersection of the chain with N :

$$0 \subseteq N \cap M_1 \subseteq N \cap M_2 \subseteq \dots$$

and its projection to M/N :

$$N \subseteq M_1 + N \subseteq M_2 + N \subseteq \dots$$

Now by assumption, there exists $j \in \mathbb{N}$ such that $N \cap M_j = N \cap M_{j+1}$ and $M_j + N = M_{j+1} + N$, then

$$0 = \frac{N \cap M_{j+1}}{N \cap M_j} = \frac{N \cap M_{j+1}}{(N \cap M_{j+1}) \cap M_j} \cong \frac{M_j + (N \cap M_{j+1})}{M_j}$$

and

$$0 = \frac{N + M_{j+1}}{N + M_j} = \frac{N + M_j + M_{j+1}}{N + M_j} \cong \frac{M_{j+1}}{M_{j+1} \cap (N + M_j)} = \frac{M_{j+1}}{(M_{j+1} \cap N) + M_j},$$

where we have used the Second Isomorphism Theorem and the modular law. Hence $M_j = M_{j+1}$. An analogous argument works for the Artinian case. $\square$

Corollary 6.11. *If $M_1, \dots, M_n$, for some $n \in \mathbb{N}$, are Noetherian (respectively Artinian) R -modules, then $M_1 \oplus \dots \oplus M_n$ is a Noetherian (respectively Artinian) R -module.*

Proof. It suffices to prove the statement for the direct sum of two Noetherian (respectively Artinian) R -modules M and N . This follows from the above theorem, applied the the submodule $M \oplus 0 \cong M$ and quotient module $(M \oplus N)/(M \oplus 0) \cong N$. $\square$

6.2. Noetherian and Artinian rings.

Definition 6.12. A ring R is called a *left Noetherian* (respectively *Artinian*) ring if R regarded as a left R -module is Noetherian (respectively Artinian).

Right Noetherian (respectively Artinian) rings are defined similarly.

Throughout, unless otherwise stated, by a Noetherian (respectively Artinian) ring, we mean a left Noetherian (respectively Artinian) ring.

The reformulation of Theorems 6.5 and 6.8 for rings are as follows:

Theorem 6.13. *Let R be a ring. The following are equivalent:*

- (i) *The ring R is Noetherian (respectively Artinian).*
- (ii) *Let A be any left ideal of R . Then A (respectively R/A) is finitely generated (respectively cogenerated).*
- (iii) *Every non-empty set S of left ideals of R has a maximal (respectively minimal) element.*

Recall that if R is a vector space over F , we can view elements of F as elements of R via multiplication with the unity; i.e. $\lambda \cdot 1$ for $\lambda \in F$. Hence we have the following examples and non-examples of Noetherian and Artinian rings.

Example 6.14. Let V be an n -dimensional vector space over a field F . Then V is both Noetherian and Artinian. For, if W is a proper subspace of V , then $\dim W < \dim V = n$. Thus any properly ascending (or descending) chain of subspaces cannot have more than $n + 1$ terms.

Example 6.15. A ring A is an *algebra* over a field F if A is an F -vector space such that

$$\lambda(ab) = a(\lambda b) = (\lambda a)b \quad \text{for all } a, b \in A \text{ and } \lambda \in F.$$

Let A be a finite-dimensional algebra with unity over a field F . Then A as a ring is both left and right Noetherian as well as Artinian. To see this, let $\dim_F(A) = n$. If we observe that each left or right ideal is a subspace of A over F , it follows that any properly ascending (or descending) chain cannot contain more than $n + 1$ terms.

(i) Let G be a group and F a field. Then the *group algebra* $F[G]$ is the algebra

$$F[G] = \left\{ \sum_{g \in G} \alpha_g g \mid \alpha_g \in F \right\}$$

over F , with multiplication given by

$$\left(\sum_{g \in G} \alpha_g g \right) \left(\sum_{h \in G} \beta_h h \right) = \sum_{g, h \in G} \alpha_g \beta_h gh.$$

If G is a finite group and F a field, then the group algebra $F[G]$ is both a Noetherian and an Artinian ring.

(ii) The $n \times n$ matrix ring $M_n(F)$ over a field F is an algebra, and thus also a Noetherian and Artinian ring, since $\dim(M_n(F)) = n^2 < \infty$.

Example 6.16. Let $R = F[x]$ be a polynomial ring over a field F in x . As we will later see, the ring $F[x]$ is a Noetherian ring. However $F[x]$ is not Artinian, for there exists a properly descending chain of ideals in R , namely,

$$R \supseteq Rx \supseteq Rx^2 \supseteq \cdots.$$

Yet, for A a non-zero ideal in R , the proper homomorphic image R/A is Artinian. Indeed, since R is a PID, we have $A = (p(x))$, for $p(x) \in F[x]$. Let $p(x) = a_0 + a_1x + \cdots + a_nx^n$. Then $F[x]/(p(x))$ is a finite-dimensional vector space over F with a basis $\{\bar{1}, \bar{x}, \dots, \bar{x}^{n-1}\}$, where we write $\bar{f(x)}$ for the coset $f(x) + A$. Hence, by Example 6.14, we have $R/A = F[x]/(p(x))$ is an Artinian ring.

Example 6.17. Let p be a prime number, and let

$$R = \left\{ \frac{m}{p^n} \in \mathbb{Q} \mid n \in \mathbb{N}, m \in \{0, 1, \dots, p^n - 1\} \text{ with } (m, p) = 1 \right\}$$

be the ring where addition is modulo $\mathbb{Z}$, and multiplication is trivial; that is, $ab = 0$ for all $a, b \in R$. Then

(i) each proper ideal in R is of the form

$$I_k = \left\{ \frac{1}{p^k}, \frac{2}{p^k}, \dots, \frac{p^k - 1}{p^k}, 0 \right\} \quad \text{for some } k \in \mathbb{N};$$

(ii) R is Artinian but not Noetherian.

Solution. (i) Let $I \neq (0)$ be any proper ideal of R , and let k be the smallest positive integer such that for some $m \in \mathbb{N}$, we have $m/p^k \notin I$, and hence $1/p^k \notin I$, else

$$\frac{1}{p^k} + \cdots + \frac{1}{p^k} = \frac{m}{p^k} \in I.$$

Consider n/p^i with $i \geq k$ and $(n, p) = 1$. We assert that $n/p^i \notin I$. Suppose on the contrary that $n/p^k \in I$. Since $(n, p) = 1$, we can find integers a and b such that $an + bp = 1$. Then from n/p^k , $1/p^{k-1} \in I$, we have that na/p^k (reduced modulo whole numbers) and bp/p^k (reduced modulo whole numbers) lie in I . Hence $1/p^k \in I$, a contradiction. Thus $n/p^k \notin I$, and hence $n/p^i \notin I$ for all $i \geq k$ and $(n, p) = 1$. Indeed, as otherwise

$$\frac{n}{p^i} + \frac{p^{i-k}}{p^i} + \frac{n}{p^i} = \frac{n}{p^k} \in I.$$

Hence,

$$I = \left\{ \frac{1}{p^{k-1}}, \frac{2}{p^{k-1}}, \dots, \frac{p^{k-1} - 1}{p^{k-1}}, 0 \right\}.$$

This ideal is denoted by I_{k-1} .

(ii) Since each ideal contains a finite number of elements, each descending chain of ideals must be finite. Hence R is Artinian.

Clearly, the chain

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$$

is an infinite properly ascending chain of left ideals, showing that R is not Noetherian. Note that although each ideal $I \neq R$ is finite and, hence, finitely generated, the ring R itself is not finitely generated.

Note that the condition of being Noetherian affects submodules, i.e. ideals of a ring R (when R is viewed as an R -module), hence restrictions affecting ideals do not yield restrictions concerning a general subring, as seen below.

Example 6.18. A subring of a Noetherian (respectively Artinian) ring need not be Noetherian (respectively Artinian). Indeed, for the Artinian case the ring of rational numbers $\mathbb{Q}$ is an Artinian ring, but its subring $\mathbb{Z}$ is not an Artinian ring.

For the Noetherian case, the ring $M_2(\mathbb{Q})$ is a Noetherian ring (compare Example 6.14), but its subring

$$R := \begin{pmatrix} \mathbb{Z} & \mathbb{Q} \\ 0 & \mathbb{Q} \end{pmatrix}$$

is not Noetherian - that is, not left Noetherian. Indeed, write

$$R = \begin{pmatrix} \mathbb{Z} & \mathbb{Q} \\ 0 & \mathbb{Q} \end{pmatrix} = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a \in \mathbb{Z}, b, c \in \mathbb{Q} \right\}.$$

One can verify that for each positive integer k ,

$$A_k = \left\{ \begin{pmatrix} 0 & m/2^k \\ 0 & 0 \end{pmatrix} \mid m \in \mathbb{Z} \right\}$$

is a left ideal of R and $A_k \subsetneq A_{k+1}$. Thus, there exists an infinite properly ascending chain of left ideals of R , namely,

$$A_1 \subset A_2 \subset A_3 \subset \dots$$

Hence R is not left Noetherian.

[However R is right Noetherian. Indeed, let

$$A = \begin{pmatrix} 0 & 0 \\ 0 & \mathbb{Q} \end{pmatrix},$$

which is a right ideal of R . Then

$$R/A \cong \begin{pmatrix} \mathbb{Z} & \mathbb{Q} \\ 0 & 0 \end{pmatrix} =: M$$

as right R -modules. Now it can be checked that non-zero right R -submodules of M are

$$\begin{pmatrix} (n) & \mathbb{Q} \\ 0 & 0 \end{pmatrix} \quad \text{where } n \text{ is a fixed integer.}$$

Hence, any ascending chain of (right) R -submodules of the R -module M gives rise to an ascending chain of ideals in $\mathbb{Z}$. It then follows that $R/A \cong M$ is right Noetherian. Also A as

a right R -module has no proper submodules. Thus A is a right Noetherian R -module. Then by Theorem 6.10, we have that R is right Noetherian.]

Theorem 6.19. *Let R_i , for $1 \leq i \leq n$, be a family of Noetherian (respectively Artinian) rings. Then their direct sum $R = \bigoplus_{i=1}^n R_i$ is again Noetherian (respectively Artinian).*

Proof. This is just Corollary 6.11, where we consider each R_i as an R -module.

Alternatively: We know that each left ideal A of R is of the form $A_1 \oplus \cdots \oplus A_n$, where A_i are left ideals in R_i . So if a left ideal $B = B_1 \oplus \cdots \oplus B_n$ of R is such that $A \subseteq B$, then it is clear that $A_i \subseteq B_i$, for $1 \leq i \leq n$. Hence, any properly ascending (respectively descending) chain of left ideals in R must be finite because each R_i is Noetherian (respectively Artinian). $\square$

Recall that for two sets A and B , we have that

$$AB \stackrel{\text{def}}{=} \left\{ \sum_{i=1}^n a_i b_i \mid n \in \mathbb{N}, a_i \in A, b_i \in B \right\}.$$

If A and B are (left) ideals in a ring R , then AB is again a (left) ideal.

Let R be a commutative ring. Recall that an ideal $P \neq R$ is *prime* if whenever $a, b \in R$ and $ab \in P$, then $a \in P$ or $b \in P$. The definition of a prime ideal can be extended to non-commutative rings, as follows:

Definition 6.20. Let R be a ring. Then an ideal $P \neq R$ is *prime* if whenever A and B are ideals in R such that $AB \subseteq P$, then $A \subseteq P$ or $B \subseteq P$.

Certainly we have that (0) is a prime ideal in an integral domain. The result below gives more examples in the case when the ring is not necessarily commutative.

Theorem 6.21. *Let R be a ring with unity. Then every maximal ideal is prime.*

Proof. Let M be a maximal ideal, and let A and B be ideals in R such that $AB \subseteq M$. Suppose $A \not\subseteq M$. Then $A + M = R$ from the maximality of M . Now write $1 = a + m$ for $a \in A$ and $m \in M$. Let $b \in B$. We have

$$b = ab + mb \in AB + M \subseteq M.$$

So $B \subseteq M$, proving that M is prime. $\square$

Note that the converse to the above theorem is not true, as (0) is prime but not maximal in $\mathbb{Z}$.